

SECTION-B

2. What is the Flynn's classification? Discuss SISD in computer architecture.
3. Explain the micro operations. Give the examples of logic and shift operations.
4. What are instruction formats? Explain direct and indirect address instruction with suitable example.
5. Give the introduction of Control units. Explain the micro programmed control unit.
6. Show the basic architecture of the Mobile device. Give some examples of it.
7. List the advantage of cache in computer architecture. Discuss the process of writing data into cache.

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Roll No.

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Total No. of Pages : 02

Total No. of Questions : 16

BCA (2014 to 2018) (Sem.-2)
COMPUTER SYSTEM ARCHITECTURE
Subject Code : BSBC-204
M.Code : 10053

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and a student has to attempt any FOUR questions.

SECTION-A

Answer briefly :

1. What is Interrupt Cycle?
2. What are Different types of Addressing Modes?
3. What do you understand by Cache memory?
4. Define Shift Operators and its types.
5. What does locality of reference mean?
6. What is difference between computer organization computer architecture?
7. Explain the term Handshaking.
8. Briefly explain the DMA controlled data transfer technique.
9. What is Set-Associative Mapping?
10. What do you understand by Register Stack?

SECTION-B

11. Write short notes on :
 - a) Layered Approach Architecture
 - b) Replacement Algorithms
12. What is meant by SISD, SIMD and MIMD architecture? Differentiate SIMD and MIMD architectures.
13. What do you mean by addressing mode? What are various types of addressing modes and their advantages and disadvantages?
14. Explain Von Neumann Architecture in detail.
15. What is an interrupt in computer organisation? Discuss interrupt types and interrupt cycle.
16. What is difference between synchronous and asynchronous data transfer?

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SECTION-B

2. What is control unit? Explain the micro programmed control unit.
3.
 - a) Draw the diagram of a JK master flip-flop and explain its operation.
 - b) Show how AND, OR, NOT gates can be realized from NAND gates?
4. Compare and contrast the features of RISC and CISC.
5. What is DMA data transfer? In what circumstances this scheme of data transfer is employed? What are burst mode and cycle stealing mode in DMA?
6. Solve the following function to SOP and POS forms using 5 variable Karnaugh map
$$F = \sum m (2, 3, 10, 11, 12, 13, 16, 17, 18, 19, 20, 21, 26, 27)$$
7. Briefly discuss the following I/O schemes :
 - a) Programmed I/O
 - b) Interrupt initiated I/O

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BCA (Sem – 2)
COMPUTER SYSTEM ARCHITECTURE

Subject Code : UGCA-1908

M.Code : 77416

Date of Examination : 06-07-22

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY** consisting of **TEN** questions carrying **TWO** marks each.
2. **SECTION-B** contains **SIX** questions carrying **TEN** marks each and students have to attempt any **FOUR** questions.

SECTION-A

Q1. Write briefly :

- Define NOR gate.
- What is meant by Truth Table?
- What is POS form?
- What are K-maps?
- What is meant by full adder?
- Name two uses of demultiplexer.
- Define an Encoder.
- What is CISC?
- What is meant by register transfer?
- Differentiate Control bus and Address bus.

SECTION-B

Q2. Generate AND, OR and NOT gates using NOR gates.

Q3. a) Discuss the working of Full Subtractor Circuit.

b) What is meant by Decoder? Explain.

- Q4. What is meant by SR Flip Flop? Explain the race-around condition in detail.
- Q5. Explain the Harvard Architecture.
- Q6. a) What are register reference instructions? Explain.
b) Explain the working of D-Flip Flop.
- Q7. Draw and explain the data movement among registers using common bus.

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